Epic 1: Accessibility Violation Detection System

Requirement Type: Functional Requirement

Requirement: System must automatically detect accessibility features and violations to quickly identify compliance issues without relying solely on manual inspections.

User Story 1.1: Image Capture

Story: As a facility manager, I want to upload or capture images of a bathroom, so that the system can analyze them for accessibility compliance.

Acceptance Criteria:

- **Given:** A valid image (JPEG or PNG). The image is either an upload of a floorplan or of the room
- **When:** User uploads or captures the photo
- **Then:** The system preprocesses it (resizing, lighting correction), stores it securely, and confirms successful upload

User Story 1.2: Violation Detection

Story: As a system user, I want to automatically detect accessibility violations in bathroom images, so that I can identify areas that fail compliance standards

Acceptance Criteria:

- **Given:** A preprocessed image
- **When:** Computer vision model runs (for images of the room), otherwise analyze floorplan dimensions
- **Then:** Identifies and highlights non-compliant features (e.g., grab bars, sink height, door width, etc.) with at least 80% detection accuracy

User Story 1.3: Standards Mapping

Story: As a compliance officer, I want to see each detected issue mapped to a specific accessibility guidelines, so that I can understand exactly which rule is being violated.

Acceptance Criteria:

- **Given:** Detected violations
- **When:** The report/audit (summary report of feedback) is generated
- **Then:** Each violation links to its official guideline reference and provides an explanation

User Story 1.4: Actionable Feedback

Story: As a facility manager, I want to receive corrective suggestions for detected violations, so that I can fix issues efficiently.

Acceptance Criteria:

- **Given:** Identified violations
- **When:** The system generates the report/audit (summary report of feedback)
- **Then:** Each violation includes annotated visuals and corrective recommendations with example measurements (exportable as a PDF).

Epic 2: Reporting and Dashboard

Requirement Type: Functional Requirement

Requirement: System must present inspection results and insights clearly and accessibly.

User Story 2.1: Report Generation

Story: As a facility manager, I want to generate a compliance report so that I can review, save, or share inspection results.

Acceptance Criteria:

- **Given:** Available analysis results
- **When:** The user requests (to view) a summary
- **Then:** The system generates a downloadable report including compliance percentage, violations, and timestamps, filterable by date or type.

User Story 2.2: Dashboard Visualization

Story: As a facility manager, I want to view a dashboard of inspection results, so that I can track compliance trends and common issues over time.

Acceptance Criteria:

- **Given:** Multiple inspection results
- **When:** The user logs-in to their dashboard
- **Then:** The system displays updated charts, compliance trends, and a ranked list of most frequent violations.

Epic 3: Model Training and Continuous Improvement

Requirement Type: Functional Requirement
Requirement: System must continuously improve detection accuracy through retraining.

User Story 3.1: Data Labeling

Story: As an administrator, I want to review and label inspection images, so that the system can use them to improve model accuracy.
Acceptance Criteria:

- **Given:** New inspection images
- **When:** The admin labels or corrects model outputs
- **Then:** The data is added to the version-controlled training dataset

User Story 3.2: Model Retraining

Story: As a developer, I want to retrain the model with new data, so that detection performance improves over time.
Acceptance Criteria:

- **Given:** Updated labeled data
- **When:** Retraining is triggered (manually)
- **Then:** The model retrains, validates accuracy against benchmarks, and deploys only if accuracy meets or exceeds baseline.

Epic 4: User Management and Access Control

Requirement Type: Functional Requirement
Requirement: System must allow secure access for different stakeholder roles.

User Story 4.1: Role-Based Access

Story: As an administrator, I want to define different access levels for users, so that each stakeholder can only access data relevant to their role.
Acceptance Criteria:

- **Given:** Authenticated session
- **When:** A user logs-in

- **Then:** Access permissions are enforced (e.g., inspectors upload, managers view dashboards, regulators see statistics, etc.), and unauthorized access is denied (can use the system, but it doesn't log/save reports)

Epic 5: Scalability

Requirement Type: Non-functional Requirement
Requirement: System should support high concurrency and performance scalability.

User Story 5.1: Scalability and Performance

Story: As a system architect, I want to ensure the platform can process many analyses concurrently, so that performance remains stable under heavy load.

Acceptance Criteria:

- **Given:** Simultaneous users
- **When:** Users upload images and trigger analysis
- **Then:** The system maintains average processing time under 10 seconds and uptime above 99%

Epic 6: Compliance, Security, and Ethics

Requirement Type: Non-Functional Requirement
Requirement: System must ensure data privacy, security, and ethical AI operation.

User Story 6.1: Data Privacy and Security

Story: As a data protection officer, I want to ensure uploaded data is secure, so that users' privacy and regulatory compliance are maintained.

Acceptance Criteria:

- **Given:** Uploaded images
- **When:** Stored or transmitted
- **Then:** All sensitive data is encrypted (AES-256 at rest, HTTPS in transit), PII (Personally Identifiable Information) is blurred or removed.

User Story 6.2: Ethical AI Transparency

Story: As a user, I want to understand how confident the system is in its results, so that I can make informed decisions about AI-generated findings.

Acceptance Criteria:

- **Given:** AI-generated results
- **When:** The system displays violation details
- **Then:** Each detection includes a disclaimer explaining limitations, with an option to dispute or provide feedback.

Epic 7: Usability and Accessibility

Requirement Type: Non-Functional Requirement

Requirement: System interface must be accessible, intuitive, and inclusive for all users.

User Story 7.1: Accessible Interface

Story: As a user requiring accessibility, I want to interact with the platform easily, so that I can access all system features independently.

Acceptance Criteria:

- **Given:** A user accessing the platform
- **When:** They navigate using keyboard or screen reader
- **Then:** The interface meets standards with high contrast, text scaling, and full keyboard accessibility. Icons are intuitive and actions require minimal clicks/steps.

User Story 7.2: User Experience

Story: As a new user, I want to complete inspections quickly and intuitively, so that I can get actionable insights without training.

Acceptance Criteria:

- **Given:** The user workflow (upload → analysis → report)
- **When:** A user completes a session
- **Then:** Average task completion time is under 3 minutes, satisfaction ≥ 85% in testing, and onboarding guidance is provided.

Product Backlog

Backlog ID	Epic/Story Title	User Story	Priority	Story Points	Acceptance Criteria Example	Sprint Target	Dependencies	Risks & Mitigations
EPIC-1	Epic 1: Accessibility Violation Detection System		High			Sprint 1		Core to project, no major dependencies
STORY-1.1	Image Capture	As a facility manager, I want to upload or capture images of a bathroom, so that the system can analyze them for accessibility compliance.	High	3	Given: A valid image (JPEG or PNG). The image is either an upload of a floorplan or of the room When: User uploads or captures the photo Then: The system preprocesses it (resizing, lighting correction), stores it securely, and confirms successful upload	Sprint 1	Basic UI framework	Risk: File upload issues Mitigation: Test file handling early
STORY-1.2	Violation Detection	As a system user, I want to automatically detect accessibility violations in bathroom images, so that I can identify	High	8	Given: A preprocessed image When: Computer vision model runs	Sprint 1	Requires STORY-1.1	Risk: Model underperforms Mitigation: Use pretrained CV

		non-compliant areas.			(for images of the room), otherwise analyze floorplan dimensions Then: Identifies and highlights non-compliant features (e.g., grab bars, sink height, door width, etc.) with at least 80% detection accuracy			models and synthetic data.
STORY-1.3	Standards Mapping	As a compliance officer, I want detected issues mapped to accessibility guidelines, so that I understand which rule is violated.	Medium	5	Given: Detected violations When: The report/audit (summary report of feedback) is generated Then: Each violation links to its official guideline reference and provides an explanation	Sprint 1	Requires STORY-1.2	Risk: Complex guideline mapping Mitigation: Use structured rule database.
STORY-1.4	Actionable Feedback	As a facility manager, I want corrective suggestions for violations, so that I can fix issues efficiently.	Medium	5	Given: Identified violations When: The system generates the report/audit (summary report of feedback)	Sprint 1	Requires STORY-1.3	Risk: Suggestion accuracy Mitigation: Manually verify initial examples.

					Then: Each violation includes annotated visuals and corrective recommendations with example measurements (exportable as a PDF).			
EPIC-2	Epic 2: Reporting & Dashboard		High			Sprint 1	Depends on EPIC-1 completion	Risk: Data presentation delays Mitigation: Use charting libraries (e.g., Chart.js).
STORY-2.1	Report Generation	As a facility manager, I want to generate a compliance report, so that I can review and share inspection results.	High	5	Given: Available analysis results When: The user requests (to view) a summary Then: The system generates a downloadable report including compliance percentage, violations, and timestamps, filterable by date or type.	Sprint 1	Depends on STORY-1.4	Risk: PDF export issues Mitigation: Use proven libraries (jsPDF).
STORY-2.2	Dashboard Visualization	As a facility manager, I want to view a	Medium	8	Given: Multiple inspection results	Sprint 2	Requires STORY-2.1	Risk: Chart rendering errors

		dashboard, so that I can track compliance trends and common issues.			When: The user logs-in to their dashboard Then: The system displays updated charts, compliance trends, and a ranked list of most frequent violations.			Mitigation: Prototype early.
EPIC-3	Epic 3: Model Training & Continuous Improvement		Medium			Sprint 2	Relies on data from EPIC-1	Risk: Time limits for retraining Mitigation: Simulate retraining with small dataset.
STORY-3.1	Data Labeling	As an admin, I want to label inspection images, so that the system can improve model accuracy.	Medium	5	Given: New inspection images When: The admin labels or corrects model outputs Then: The data is added to the version-controlled training dataset	Sprint 2	Requires STORY-1.2	Risk: Limited labeled data Mitigation: Use open datasets or synthetic images.
STORY-3.2	Model Retraining	As an AI engineer, I want to retrain the model, so that detection performance improves over time.	Medium	8	Given: Updated labeled data When: Retraining is triggered (manually) Then: The model	Sprint 2	Requires STORY-3.1	Risk: Overfitting Mitigation: Maintain validation set.

					retrains, validates accuracy against benchmarks, and deploys only if accuracy meets or exceeds baseline.			
EPIC-4	Epic 4: User Management & Access Control		Medium			Sprint 1	Independent	Risk: Misconfigured roles Mitigation: Implement RBAC early.
STORY-4.1	Role-Based Access	As an admin, I want to define user access levels, so that each stakeholder can only access permitted data.	Medium	5	Given: Authenticated session When: A user logs-in Then: Access permissions are enforced (e.g., inspectors upload, managers view dashboards, regulators see statistics, etc.), and unauthorized access is denied (can use the system, but it doesn't log/save reports)	Sprint 1	Authentication setup	Risk: Security gaps Mitigation: Use tested auth libraries.
EPIC-5	Epic 5:		Low			Sprint 2	Depends on	Risk: Limited

	Scalability & Performance						core system readiness	hosting performance Mitigation: Optimize batch size and use cloud GPU.
STORY-5.1	Scalability and Performance	As a system architect, I want the platform to handle many analyses concurrently, so that performance remains stable.	Low	5	Given: Simultaneous users When: Users upload images and trigger analysis Then: The system maintains average processing time under 10 seconds and uptime above 99%	Sprint 2	Requires core detection system	Risk: Resource limits → Mitigation: Use parallel processing.
EPIC-6	Epic 6: Compliance, Security & Ethics		High			Sprint 2	Applies across all epics	Risk: Privacy noncompliance Mitigation: Conduct privacy review.
STORY-6.1	Data Privacy and Security	As a data protection officer, I want all uploaded data to be secure, so that user privacy and legal compliance are ensured.	High	5	Given: Uploaded images When: Stored or transmitted Then: All sensitive data is encrypted (AES-256 at rest,	Sprint 2	Applies to all image features	Risk: Data leaks Mitigation: Enforce encryption and limited access.

					HTTPS in transit), PII (Personally Identifiable Information) is blurred or removed.			
STORY-6.2	Ethical AI Transparency	As a user, I want to see AI confidence levels, so that I can trust and verify the results.	Medium	3	Given: AI-generated results When: The system displays violation details Then: Each detection includes a disclaimer explaining limitations, with an option to dispute or provide feedback.	Sprint 2	Requires detection module	Risk: Misinterpretation Mitigation: Add clear disclaimers.
EPIC-7	Epic 7: Usability & Accessibility		Medium			Sprint 2	Final UI polish	Risk: Accessibility issues Mitigation: Follow WCAG 2.1 AA guidelines.
STORY-7.1	Accessible Interface	As a user with a disability, I want the platform to be accessible, so that I can use all features independently.	Medium	5	Given: A user accessing the platform When: They navigate using keyboard or screen reader	Sprint 2	Requires UI completion	Risk: Accessibility gaps Mitigation: Test with screen readers.

					Then: The interface meets standards with high contrast, text scaling, and full keyboard accessibility. Icons are intuitive and actions require minimal clicks/steps.			
STORY-7.2	User Experience	As a new user, I want the interface to be intuitive, so that I can complete inspections quickly.	Medium	5	Given: The user workflow (upload → analysis → report) When: A user completes a session Then: Average task completion time is under 3 minutes, satisfaction 85% in testing, and onboarding guidance is provided.	Sprint 2	Depends on core flow readiness	Risk: UX confusion Mitigation: Add onboarding and tutorials.